

Autodesk® Scaleform®

Scaleform 3D

Scaleform 3.2

3D

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: 2012 5 9

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Autodesk Scaleform

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Scaleform® 3Di™

3D Flash

UI

Scaleform 3.2

Scaleform

3D

ActionScript

. Scaleform 3.2 ActionScript

2.0

, AS 3.0

3D AS

(_z, _zscale, _xrotation, _yrotation, _matrix3d, _perspfov)

, , HUD

UI

3D

. 3D

movieclip, button textfield

3D

Scaleform Direct Access API

C++

. API

3D Gfx::Value

movieclip, button, textfield

,

3D

3D Flash

Scaleform

3Di

Flash 10

3D

. 3Di

Scaleform

AS2

2.1

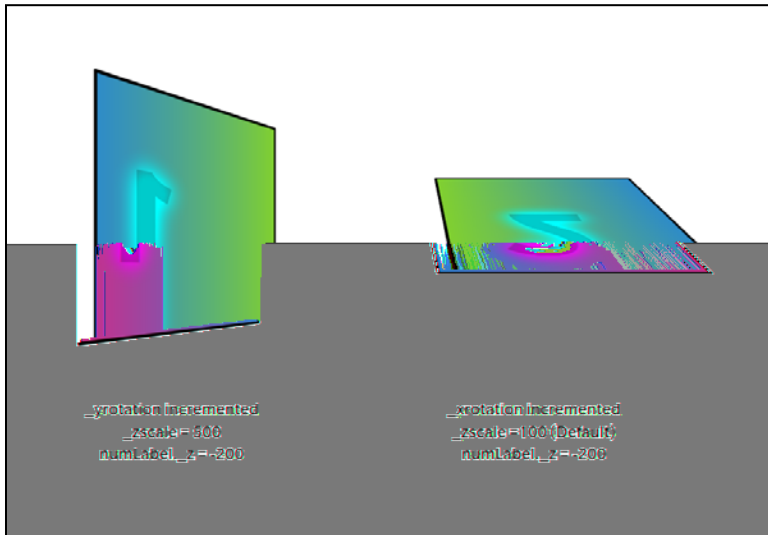
Flash 10

3D

3Di

1.

2. (Back-face culling)



2.2

Flash 10 3D , Scaleform 3Di .
2D 3D , 3D 2D .
3D . Flash 10 .

Scaleform 3.2 3D Flash 2

3.1

Flash 3D
Scaleform 3.2 3D Flash
3Di . Flash 3D
, 3D , 3D
Z() , / (UI
)
, Flash (Scaleform) Flash

movie clip
'SWF to Texture' Scaleform SDK Browser

3.2 3Di

Scaleform 3.2 , ActionScript Direct Access
API(C++) 3Di . 3Di
movie clip movie
Scaleform
Scaleform
, 3Di Scaleform UI
(back-face culling) , UI
, UI
, 3D
UI

4.1 ActionScript 2 API

Flash 3Di 2D 3D
3D

2 3D , 3D
 (x y , z , 3d), 3D
 3D 3d
 null 2D (ActionScript foo._matrix3d = null). z
 0 3D
 , Z 2D , 3D 3 x, y, z
 , 3D _zscale ActionScript

Scaleform 3D Flash 10
 , +X , +Y +Z ()
) 55 FOV +Z

3D Flash

- Movie clips
- Text fields
- Buttons

3Di ActionScript

- - Z () 0
- - Z
- 0
- - X() 0
- - Y() 0
- - 16 float (4 x 4) 3D
- NULL 3D 2D
- - (Field Of View) 0
- 180 -1
- FOV (55)

```
gfxExtensions    ActionScript    true
```

•

movie clip

. movie clip

```

    . _xrotation

```

_yrotation movie clip 3D

•

. 3Di

Y	movie clip	45
---	------------	----

```
_global.gfxExtensions = true;
mc1._yrotation = 45;
```

$$Z \qquad \qquad \qquad X$$

```
_global.gfxExtensions = true;
sq1._zscale = 500;
```

```
function rotateAboutXAxis(mc:MovieClip):Void
{
    mc._xrotation = mc._xrotation + 1;
    if (mc._xrotation > 360)
    {
        mc._xrotation = 0;
    }
}
```

```
onEnterFrame = function()
{
    rotateAboutXAxis(sq1);
}
```

3D

3D

.

```
function PixelsToTwips(iPixels):Number
{
    return iPixels*20;
}
```

```
function TranslationMatrix(tX, tY, tZ):Array
{
    var matrixC:Array = [ 1, 0, 0, 0,
                          0, 1, 0, 0,
                          0, 0, 1, 0,
                          tX,tY,tZ,1];
    return matrixC;
}
```

```
this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);
```

4.2 C++ 3Di

4x4 Matrix4x4 .

3D (,)

. Render_Matrix4x4.h .

Gfx::Value Direct Access 3Di . 3D

API . 3Di ActionScript

. Gfx_Player.h [Gfx::Value::DisplayInfo](#) .

```

void SetZ(Double z)
void SetXRotation(Double degrees)
void SetYRotation(Double degrees)
void SetZScale(Double zscale)
void SetFOV(Double fov)
void SetProjectionMatrix3D(const Matrix4F *pmat)
void SetViewMatrix3D(const Matrix3F *pmat)
bool SetMatrix3D(const Render::Matrix3F& mat)

Double GetZ() const
Double GetXRotation() const
Double GetYRotation() const
Double GetZScale() const
Double GetFOV() const
const Matrix4F* GetProjectionMatrix3D() const
const Matrix3F* GetViewMatrix3D() const
bool GetMatrix3D(Render::Matrix3F* pmat) const

```

3D

Render::TreeNode

GfX_Player.h

```
void      SetProjectionMatrix3D(const Matrix4F& m)
void      SetViewMatrix3D(const Matrix3F& m);

Ptr<GfX::Movie> pMovie = ...;
GfX::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
    GfX::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        // set perspectiveFOV to 150 degrees
        Double oldPersp = dinfo.GetFOV();
        dinfo.SetFOV(150);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
```

Direct Access API

FOV()

FOV 0

FOV

z

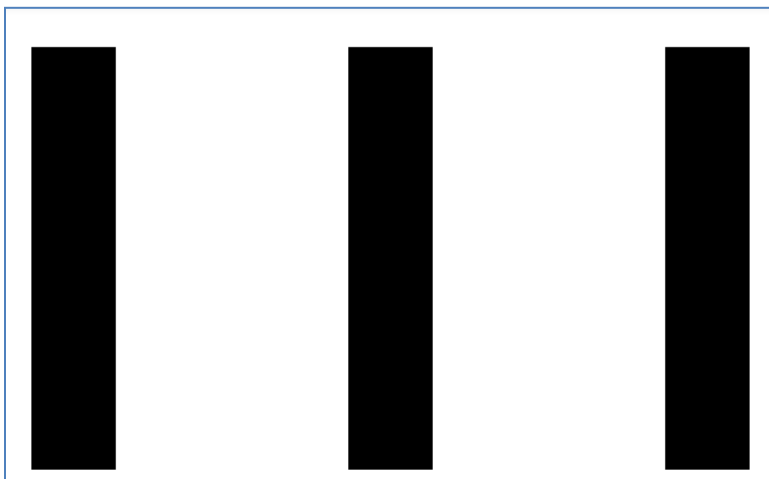
. FOV

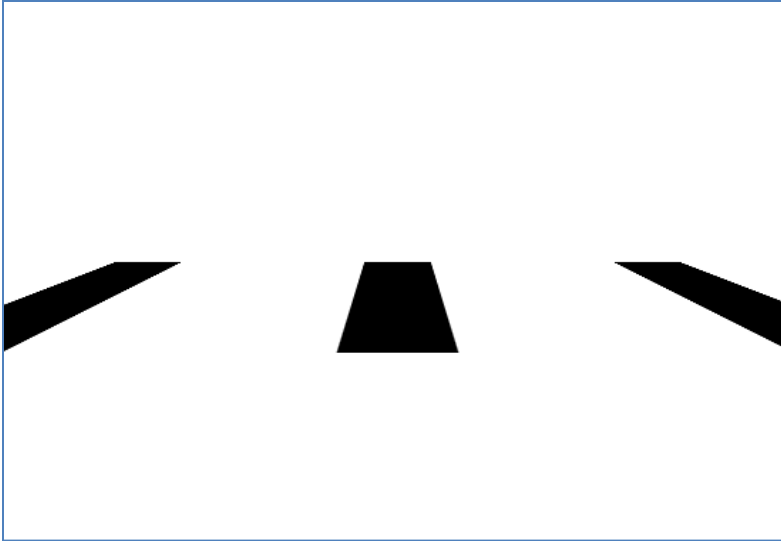
0

180

. 1

, 179





FOV (5)

. ActionScript _perspfov

-1

. C++

Direct Access API

```
// 1.    FOV    -1
Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root"); //"_root" for AS2
if (bOK)
{
    Gfx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        dinfo.SetPerspFOV(-1);
        tmpVal.SetDisplayInfo(dinfo);
    }
}

// 2.
void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)
{
    const float nearZ = 1;
    const float farZ = 100000;
```



```

float DisplayHeight = fabsf(visFrameRectInTwips.Height());
float DisplayWidth = fabsf(visFrameRectInTwips.Width());
matPersp->OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root.mc");
if (bOK)
{
    Gfx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        Matrix4F perspMat;
        MakeOrthogProj(PixelsToTwips (pMovie->GetVisibleFrameRect()), &perspMat);
        dinfo.SetProjectionMatrix3D(&perspMat);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
}

```

```

Ptr<Movie> pMovie = ...
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie->GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()), fabs(visFrameRectInTwips.Height()),
nearZ, farZ);
pMovie->SetProjectionMatrix3D(ortho);

```

5.1 AS3

ActionScript 3 PerspectiveProjection

Example:

```

import flash.display.Sprite;

var par:Sprite = new Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0, 0, 100, 100);
par.x = 100;

```

```
par.y = 100;
par.rotationX = 20;

addChild(par);

var chRed:Sprite = new Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0, 0, 100, 100);
chRed.x = 50;
chRed.y = 50;
chRed.z = 0;

par.addChild(chRed);

var pp3:PerspectiveProjection=new PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection = pp3;
```

UI

3D

. Scaleform

TinyPlayerStereo

Scaleform

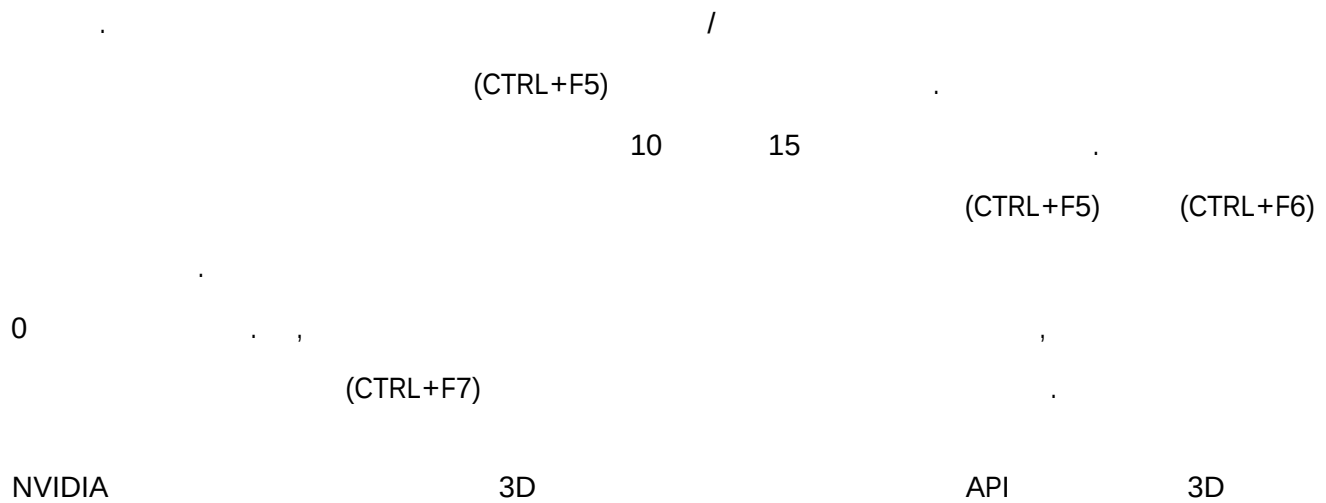
6.1 NVIDIA 3D Vision

3Di

3D

Stereoscopic 3D Test

, D3D9 Scaleform Player .
 Automatic 3D Vision Direct3D Scaleform Player
 . Scaleform Player -f
 . NVIDIA 3D
 . 3D (CTRL + T)
 .
 .
 .
 + -
 . 3D Vision ,
 .
 3D Vision IR (CTRL+F3)/(CTRL+F4)
 .
 .
 Scaleform Player .
 (CTRL+F5) (CTRL+F6) ,
 .
 .
 NVidia "advanced in-game settings
 " , "Stereoscopic 3D" "Set up stereoscopic 3D"
 "Set Keyboard Shortcuts" (CTRL+F5) (CTRL+F6)
 . (CTRL+F7) .



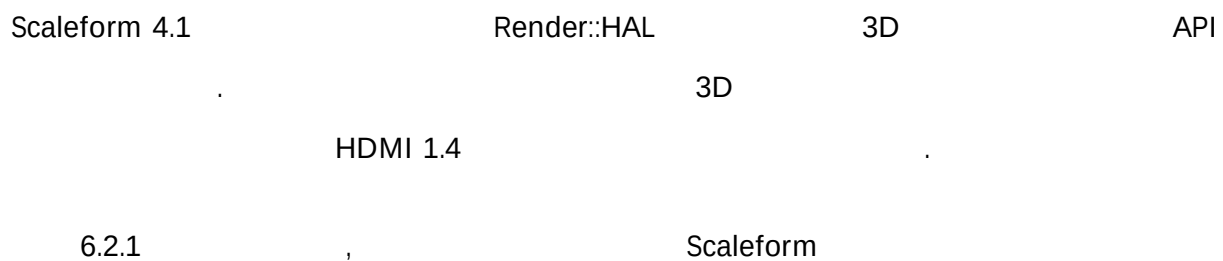
API

<http://developer.nvidia.com/object/nvapi.html>.

Designing for stereo()

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform API



Scaleform 4

SF4

Viewport Stereo

```
//  
  
View_Stereo_SplitV      = 0x40,  
View_Stereo_SplitH      = 0x80,  
View_Stereo_AnySplit    = 0xc0,  
  
void SetStereoViewport(unsigned display);
```

```
Render::StereoParams s3DInfo;  
s3DInfo.DisplayDiagInches = 46;      // 46      TV  
s3DInfo.DisplayAspectRatio = 16.f / 9.f;  
s3DInfo.Distortion = .75;           // 0      1  
s3DInfo.EyeSeparationCm = 6.4;      //      6.4cm  
pRenderHAL->SetStereoParams(s3DInfo);
```

Display

Display

Scaleform

```
pMovie->Advance(delta);  
  
pRenderer->GetHAL()->SetStereoDisplay(Render::StereoLeft);  
pMovie->Display();  
  
pRenderer->GetHAL()->SetStereoDisplay(Render::StereoRight);  
pMovie->Display();
```

Scaleform SDK

3D

Flash

(SWF, FLA)

3Di

. C:\Program Files (x86)\Scaleform\GFx SDK

4.3.Bin\Data\AS2 or AS3\Samples

